SAITEJA VULLI

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PROFESSIONAL SUMMARY

Software and Data Engineer with **2+ years of software development experience** and current experience in **AI/ML research**. I enjoy building practical systems and solving problems using **Java, Python, SQL, Spring Boot, Kafka, Spark, PostgreSQL and Docker**. My experience includes developing **backend applications, REST APIs, real-time data pipelines, ETL workflows and data-driven applications**. In my current research role, I work with **deep learning and Generative AI**, including diffusion models, VQ-VAE, transformers, CLIP, and PyTorch. I have also built end-to-end projects involving **real-time data processing, machine learning, databases, and scalable backend systems**. I am comfortable learning new technologies, working through complex technical problems and turning ideas into reliable solutions that can be used in real-world applications.

TECHNICAL SKILLS

Programming Languages:	Java, Python, SQL (PostgreSQL), Rust, JavaScript, C/C++, R, HTML/CSS
Data Engineering:	ETL Pipelines, Data Ingestion, Data Transformation, Batch & Real-Time Processing, Data Integration, Medallion Architecture (Bronze/Silver/Gold), Structured Streaming, Event-Driven Architecture
Big Data & Streaming:	Apache Spark, PySpark, Spark Structured Streaming, Apache Kafka, Apache Parquet, DataFusion
Databases:	PostgreSQL, MongoDB, SQL, Relational Databases, Database Design, Analytical SQL
Backend Development:	Spring Boot, REST APIs, FastAPI, Flask, Node.js, Microservices, API Integration, JSON
AI / Machine Learning:	PyTorch, MONAI, TensorFlow, Scikit-learn, Pandas, NumPy, Deep Learning, CNNs, Diffusion Models, VQ-VAE, Transformers, CLIP Embeddings, Model Training & Evaluation, Hyperparameter Tuning
Cloud & Infrastructure:	Google Cloud Platform (GCP), Docker, Docker Compose, Containerization
Data Analytics & Visualization:	Tableau, Streamlit, Matplotlib, Data Analysis, Data Visualization
Developer Tools:	Git, GitHub, IntelliJ IDEA, VS Code, PyCharm, pgAdmin, Postman, Google Colab, MATLAB
Software Engineering:	Data Structures & Algorithms, Object-Oriented Programming (OOP), Multithreading, CLI Development, Performance Optimization, Benchmarking, Error Handling, Scalable System Design
Architectures & Concepts:	Real-Time Data Processing, Event-Driven Systems, RESTful Architecture, Medallion Architecture, Data Pipelines, Distributed Data Processing

WORK EXPERIENCE

Rowan University
Jan 2026 – Present

Graduate Research Assistant — Deep Learning & Generative AI
Glassboro, NJ

- Conduct research on **deep learning, diffusion models, and representation learning**, with a focus on developing and evaluating generative AI approaches.
- Develop **VQ-VAE and diffusion-based pipelines using PyTorch and MONAI** for structured-to-latent representation learning and reconstruction.
- Integrate **transformers and CLIP embeddings** into experimental pipelines to improve representation alignment and model performance.
- Train and evaluate deep learning models on **GPU environments**, performing hyperparameter tuning, model validation, and performance analysis.
- Build and test experimental workflows in **Python**, analyzing model outputs and comparing results across different architectures and configurations.
- Collaborate with faculty on research experiments, architecture improvements, debugging, and evaluation of model performance.

Aagama IT Biz Solutions Pvt. Ltd.

Jan 2023 – Dec 2024

Frontend Developer

Hyderabad, India

- Developed and maintained responsive **web application interfaces** for a real estate platform using **JavaScript, CSS, Bootstrap, and Node.js**.
- Built reusable UI components and integrated **REST APIs** to support real-time data updates and application functionality.
- Worked with backend API responses and **JSON data** to display dynamic information across application pages.
- Improved application performance by reducing unnecessary re-renders and optimizing page loading and frontend behavior.
- Tested and debugged application functionality to identify UI, API integration, and performance issues.
- Collaborated with the development team to implement features, resolve bugs, and improve the overall user experience.

PROJECTS

AetherFlow — AI Research Data Platform

Python | Apache Kafka | PySpark | Spark Structured Streaming | Docker | Parquet | Streamlit

- Built an **end-to-end real-time data pipeline** for AI/ML experiment telemetry using **Apache Kafka and Spark Structured Streaming** with a **Bronze–Silver–Gold Medallion architecture**.
- Developed **PySpark ETL pipelines** to transform and store model performance, experiment tracking, and GPU utilization data in **Parquet**.
- Built an interactive **Streamlit dashboard** for monitoring experiment metrics and analyzing model and GPU performance.

Banking Fraud Detection Data Platform

Python | Apache Kafka | Apache Spark | PostgreSQL | Docker | Streamlit

- Developed a **real-time fraud detection data pipeline** using Kafka and Spark Structured Streaming to ingest and process simulated banking transactions.
- Implemented **transaction validation, rule-based fraud detection, and Dead Letter Queue (DLQ) handling**, storing processed transactions and fraud alerts in PostgreSQL.
- Created analytical **SQL views and Streamlit/Tableau dashboards** to track transaction volume, fraud rates, high-risk patterns, and fraud trends.
- Containerized the complete platform using **Docker and Docker Compose**, demonstrating a deployable event-driven data architecture.
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Parquet Compactor CLI

Rust | Polars | Clap | Indicatif | TOML | Multithreading

- Developed a high-performance **Rust CLI tool** that addresses the small-file problem by combining multiple Parquet files into optimized larger files.
- Implemented **schema validation, configurable target file sizes, dry-run mode, TOML configuration, progress tracking, and robust error handling**.
- Added **multithreading and Criterion benchmarks** to evaluate and improve processing performance and scalability.

Resume Analyzer

Java | Spring Boot | React | PostgreSQL | Apache PDFBox

- Built a full-stack **resume analysis and job-matching application** with PDF resume upload, text extraction, and match-score functionality.
- Developed **REST APIs using Spring Boot** and integrated PostgreSQL for resume data storage and **Apache PDFBox** for PDF text extraction.
- Created a responsive **React frontend** to display match scores, matched skills, missing skills, and improvement suggestions.

EDUCATION

Rowan University *Jan 2025 – Dec 2026*

Master of Science – Data Science